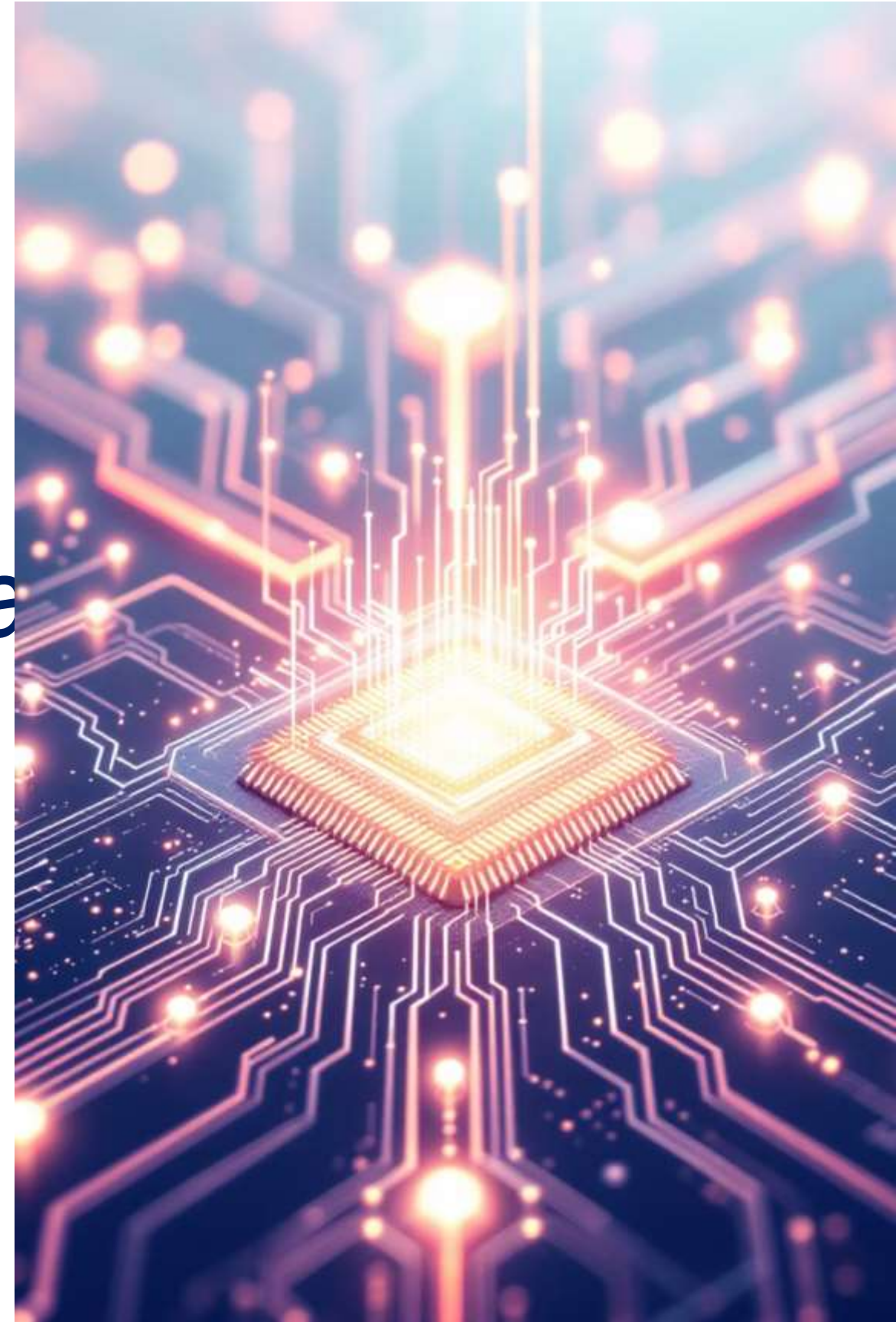


CircuitGuard

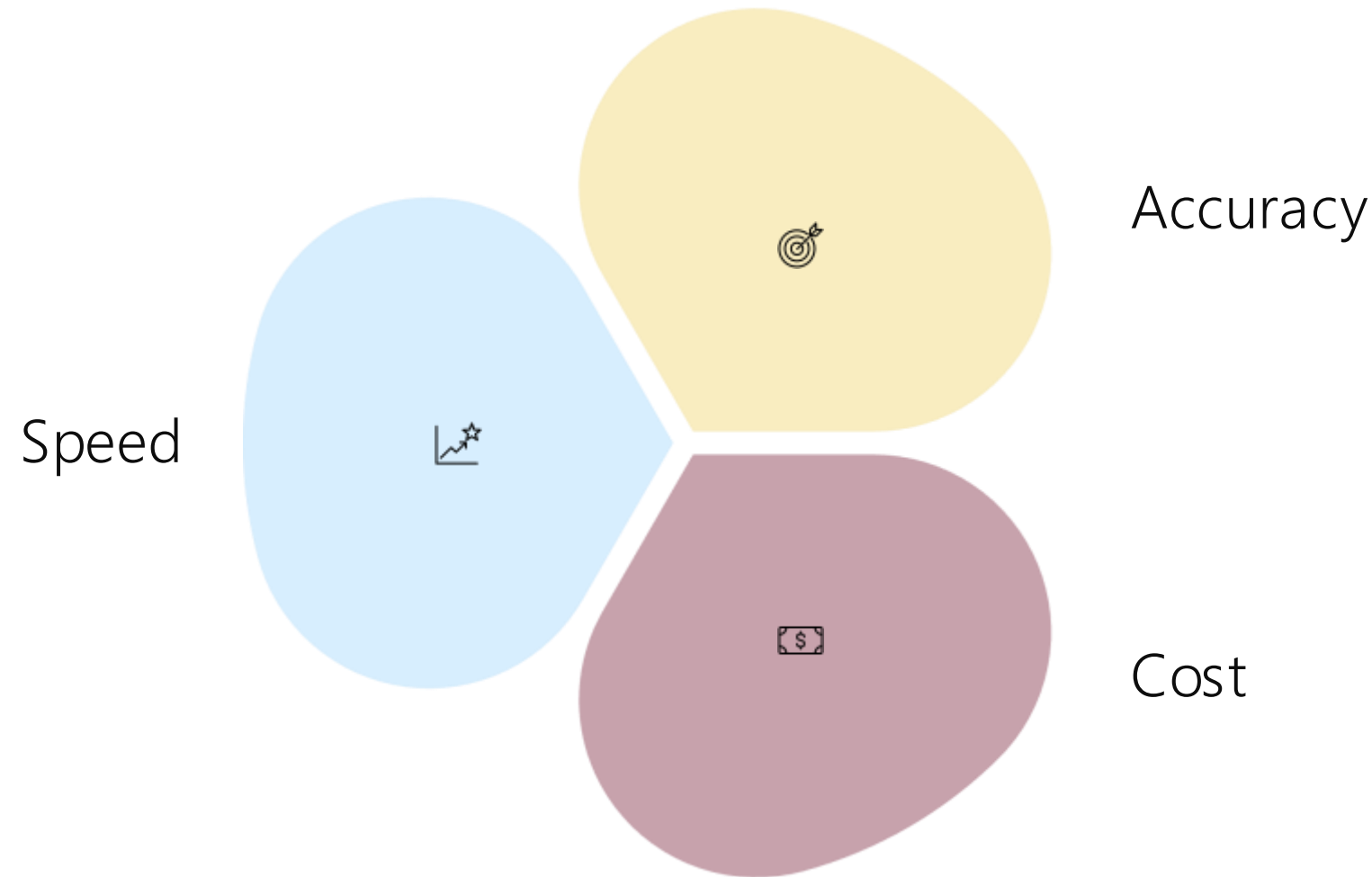
AI-Driven Automated Optical Inspection

Powered by EfficientNet-B4 & Computer Vision

Presented by : **T. Vivek**



The Industry's "Tri-Lemma"



Manufacturers face a critical choice: pick only two from **Speed, Accuracy, and Cost**.

- **Manual Inspection:** Low cost, but slow and error-prone.
- **Hardware AOI:** Fast & accurate, but expensive

Small to medium manufacturers lack viable options. Our goal: Democratize AOI with software to achieve all three.

The CircuitGuard Solution



A Hybrid Approach

We combine techniques for robust inspection:

- **Deterministic CV:** Pixel-perfect localization ("Where is it?").
- **Probabilistic AI:** Semantic classification ("What is it?").

Key Differentiator

Software-centric. No proprietary cameras or specialized hardware required.

Result

Industrial-grade inspection on consumer-grade hardware.

Project Objectives



Automate Detection

Detect six common PCB defects: Short, Open, Mouse Bite, Spur, Spurious Copper, and Missing Hole with high precision.



Ensure Precision

Use Golden Template comparison to localize defects down to the pixel level with minimal false positives.



Smart Classification

Deploy trained EfficientNet-B4 model to accurately classify defect types with high confidence.



User-Friendly Deployment

Build an intuitive web-based interface using Streamlit for single-click inspection and real-time results.

System Architecture



Input

User uploads test PCB image



Auto-Matching

System finds matching Golden Template from database



Alignment

ORB feature matching aligns test to template



Extraction

Subtraction and thresholding isolate anomalies



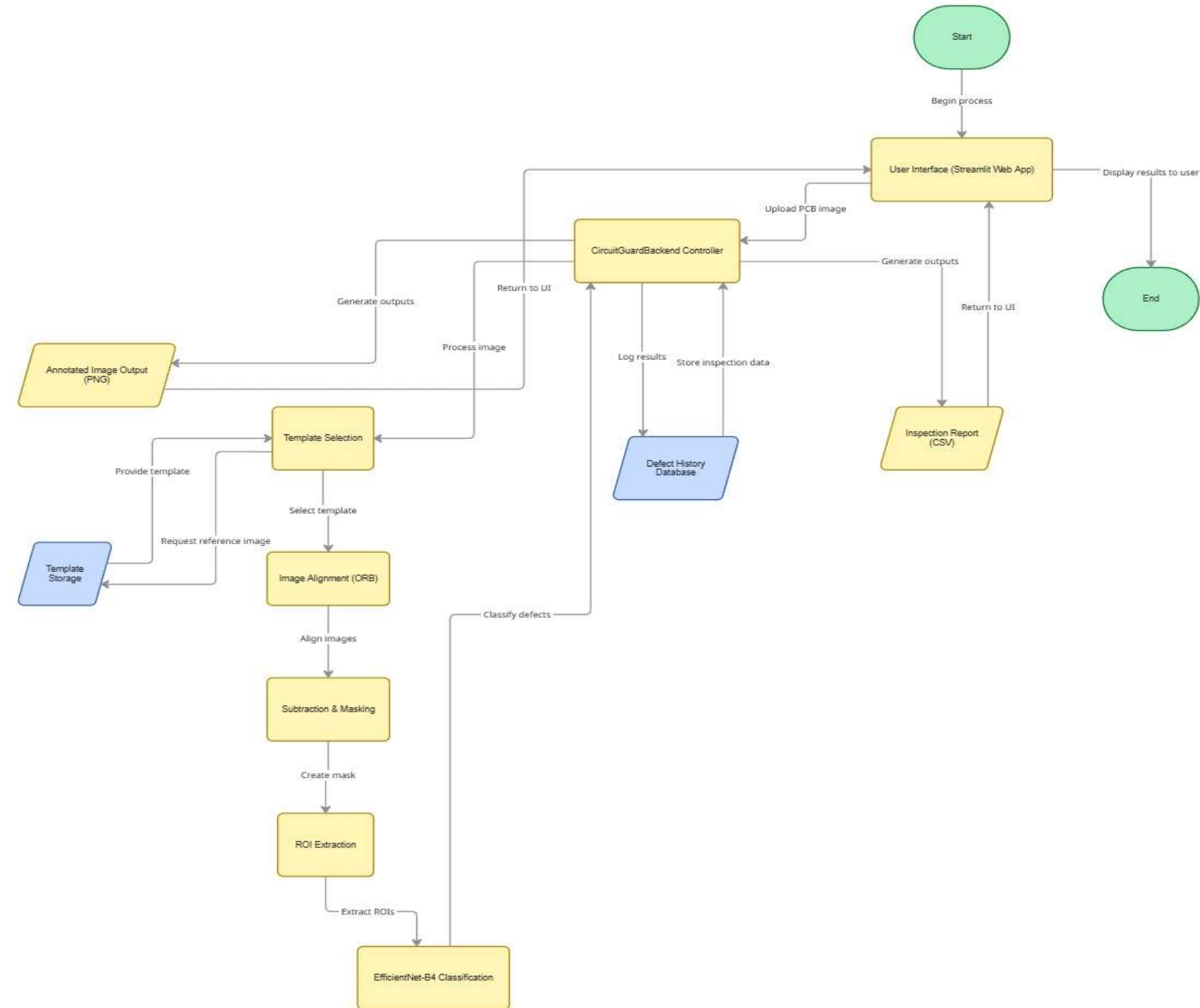
AI Analysis

EfficientNet-B4 classifies defects

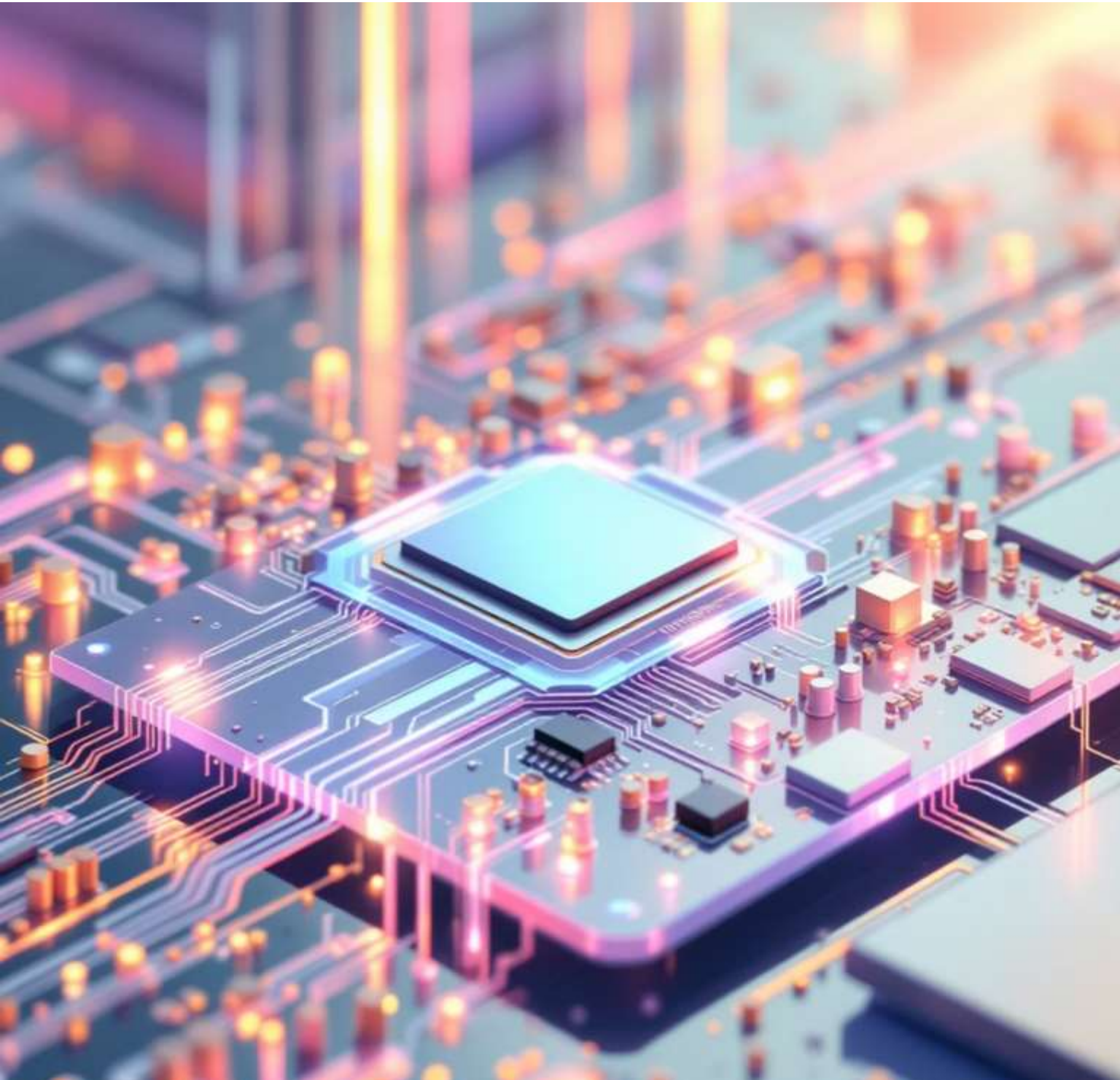


Output

Annotated image with defect report



Innovation I: The "Ghosting" Challenge



The Challenge

Simple image subtraction fails if board is rotated even 1°, causing "ghosting."

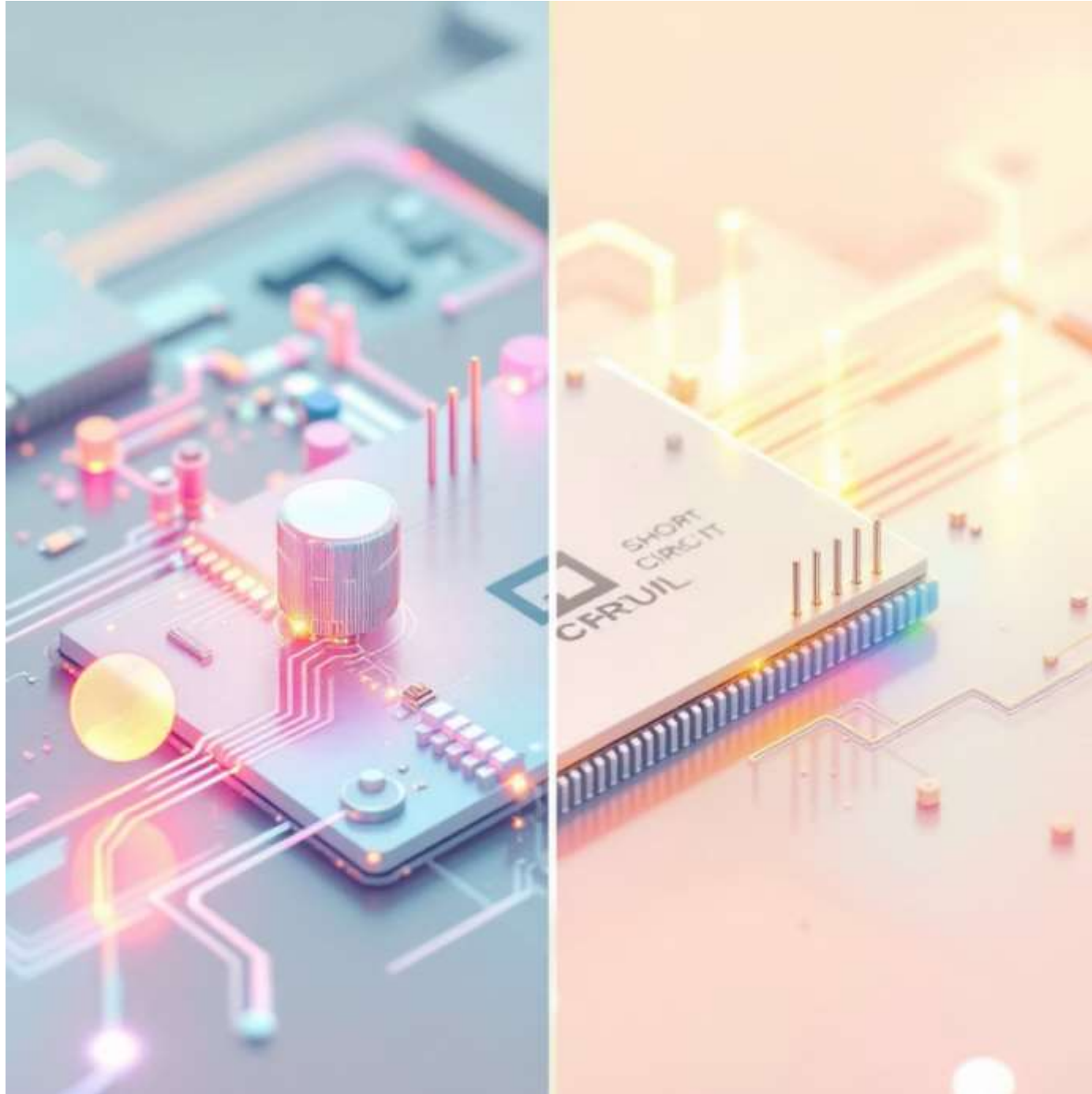
Our Solution

Auto-Template Matching & Geometric Alignment.

- **ORB Feature Matching:** Detects keypoints to perfectly align test PCB over the template with sub-pixel accuracy
- **Gaussian Blur:** Applied before subtraction to eliminate sensor noise and minor variations
- **Morphological Opening:** A digital eraser that removes dust particles while preserving real defects

Impact: Perfect pixel-accurate alignment, regardless of conveyor belt position.

Innovation II: Solving "Context Blindness"



The Problem

Standard CNNs cropped too tightly, making a "short" look like a normal wire.

The Fix

We feed the AI the **surrounding area**, not just the defect.

The Model

EfficientNet-B4 selected for high accuracy & parameter efficiency.

Training Results

Trained on 2,936 high-quality ROIs for 30 epochs, achieving improvement in classification performance.

Tech Stack



Core Language

Python 3.10+ for all development and deployment components



User Interface

Streamlit for intuitive web application deployment



Computer Vision

OpenCV for image processing and NumPy for numerical operations



Deployment

Google Colab Pro with Cloudflare Tunnel for secure access



Deep Learning

PyTorch framework with timm library for EfficientNet implementation

Key Innovations

The System Reliability Features That Makes CircuitGuard Special

1

Auto-Template Matching

Users only upload the test image—the system intelligently searches for the correct reference board in the background database. No manual template selection required, making the workflow seamless and foolproof.

2

Self-Healing Backend

If template files are missing due to system resets or storage issues, the system automatically detects the problem and re-extracts the dataset without crashing or requiring manual intervention.

3

The Confidence Gate

Implemented intelligent logic filtering where the AI must achieve >90% confidence to flag a defect. This single innovation eliminated the catastrophic "600 false positives" error entirely, ensuring reliability in production environments.

Results & Performance

Test Accuracy

95%

Final model performance across all defect classes

F1-Score

0.9392

Consistent performance across six defect categories

Inference Speed

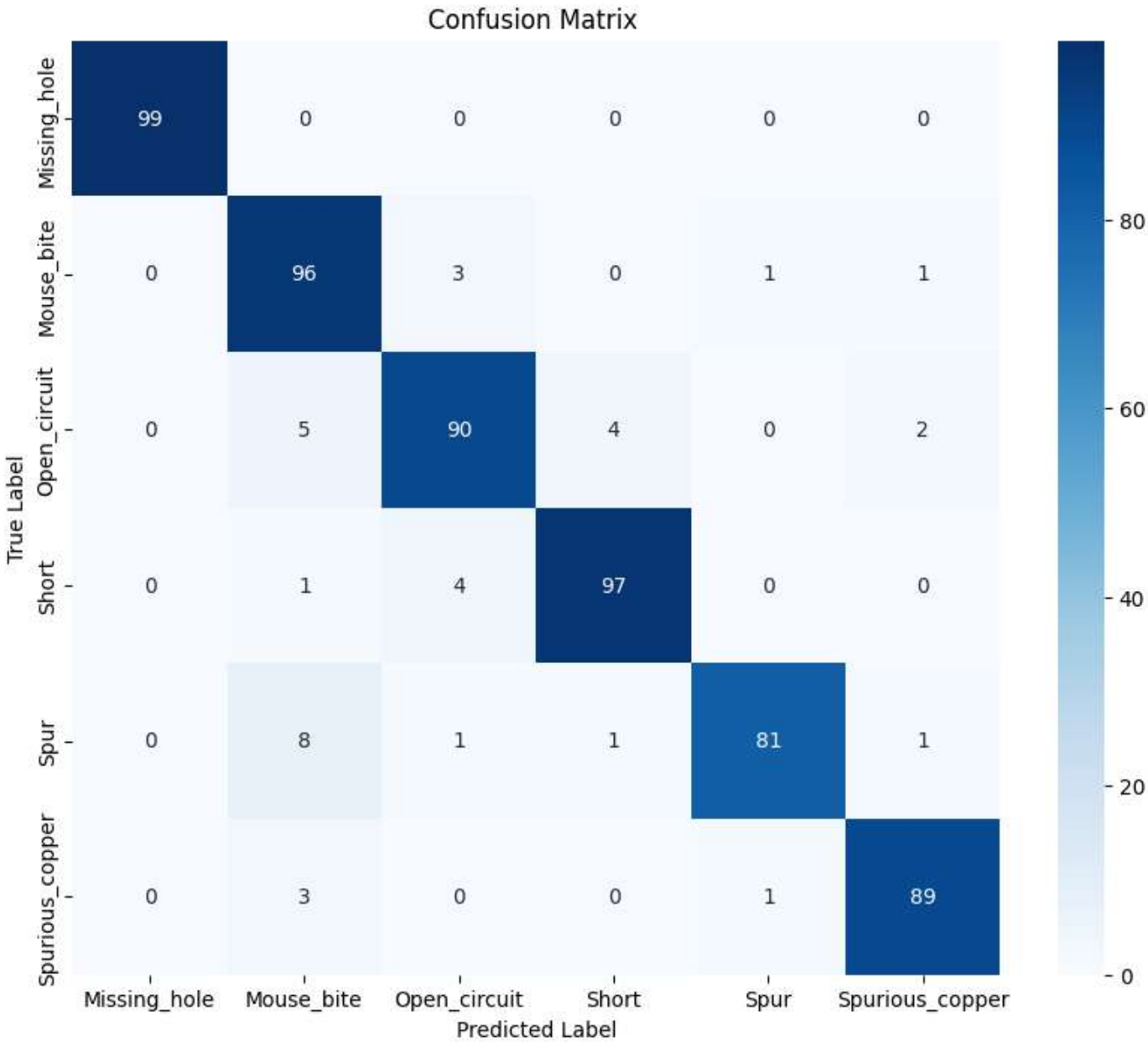
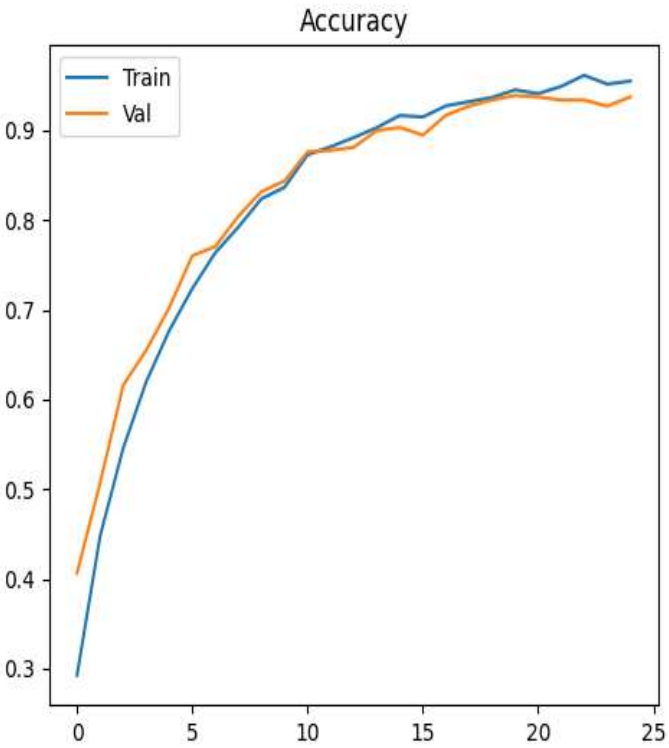
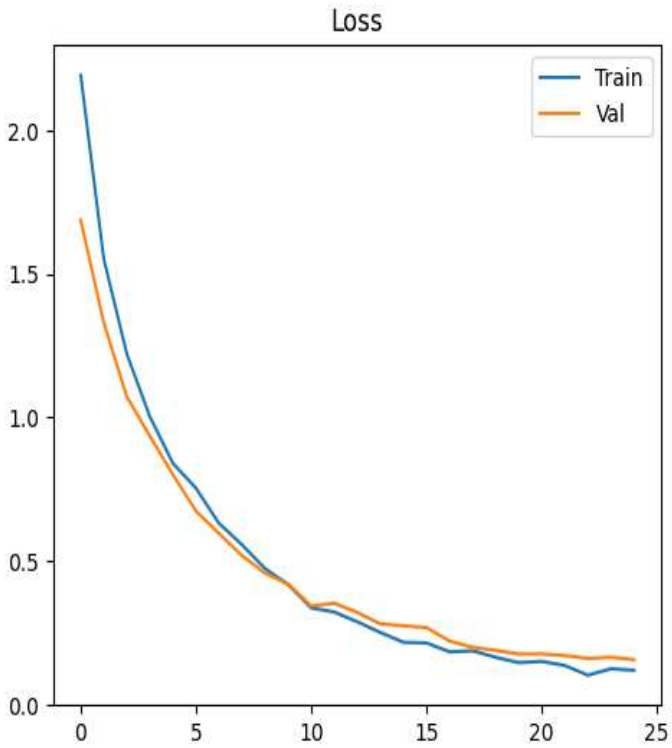
<2s

Per board processing time including visualization

False Positive Reduction

85%

Improvement over baseline approach



Conclusion & Future Directions

Project Impact

CircuitGuard successfully automates PCB inspection, dramatically reducing human error and inspection time while maintaining high accuracy. The hybrid approach proves that combining classical computer vision with modern deep learning creates robust, production-ready systems.

Future Scope

- Deploy to edge devices (Raspberry Pi, Jetson Nano) for real-time factory floor integration
- Add live camera support for continuous video stream inspection
- Expand dataset to include soldering defects and component placement errors
- Implement active learning pipeline for continuous model improvement

Thank You!